

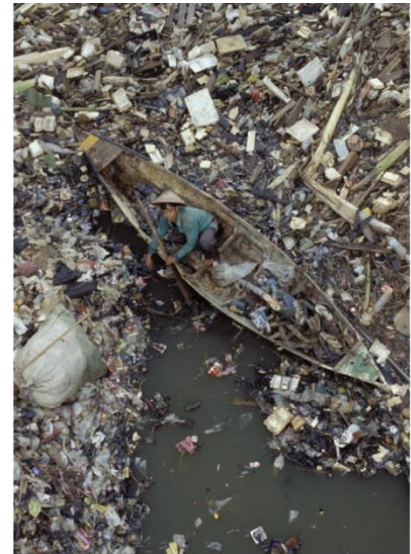
Soil Waste Management

Soil Waste Management

Solid waste management refers to the process of collecting and treating solid wastes. It also offers solutions for recycling items that do not belong to garbage or trash. As long as people have been living in settlements and residential areas, garbage or solid waste has been an issue. Waste management is all about how solid waste can be changed and used as a valuable resource.

Solid waste refers to any unwanted or discarded material we produce that is not a liquid or a gas. Solid waste can be divided into two types.

1. **Industrial solid waste:** produced by mines, farms, and industries that supply people with goods and services.
2. **Municipal solid waste (MSW):** often called garbage or trash, which consists of the combined solid waste produced by homes and workplaces. Examples include paper and cardboard, food wastes, cans, bottles, yard wastes, plastics, metals, glass and e-waste.
3. **Hazardous/toxic waste:** which threatens human health or the environment because it is poisonous, dangerously chemically reactive, corrosive, or flammable. Examples include industrial solvents, hospital medical waste, car batteries (containing lead and acids), household pesticide products, dry-cell batteries (containing mercury and cadmium), and incinerator ash.



Solid waste polluting a river

Waste Management

Solid waste can be dealt with in two ways, the first being waste management, while the second approach is waste reduction.

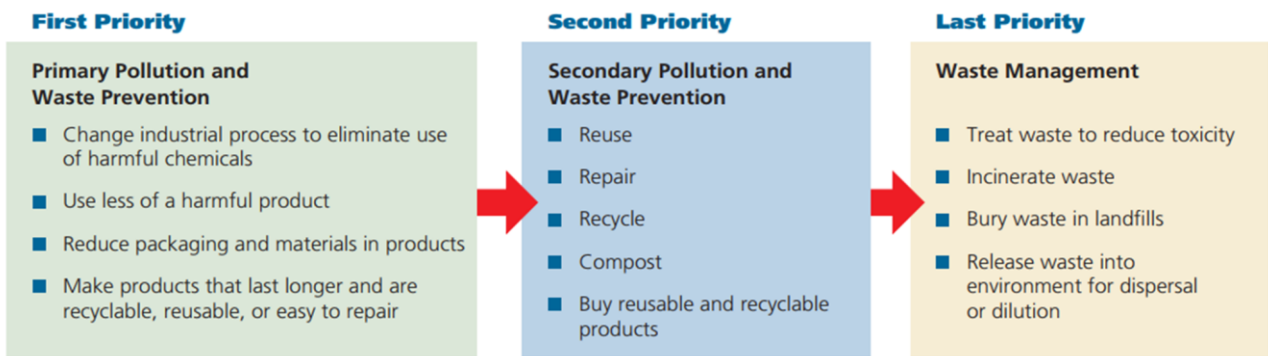
- **Waste Management** refers to controlling waste in ways so as to reduce their environmental harm without any serious efforts to reduce the amount of waste produced.
- **Waste Reduction** refers to the approach by which lesser amounts of waste and pollution is produced. So, there is greater emphasis on the reuse, recycling and composting of resources.



But there is no single solution to the solid waste problem. Most analysts call for using integrated waste management, which include a variety of coordinated strategies for both waste disposal and waste reduction.

Integrated Solid Waste Management

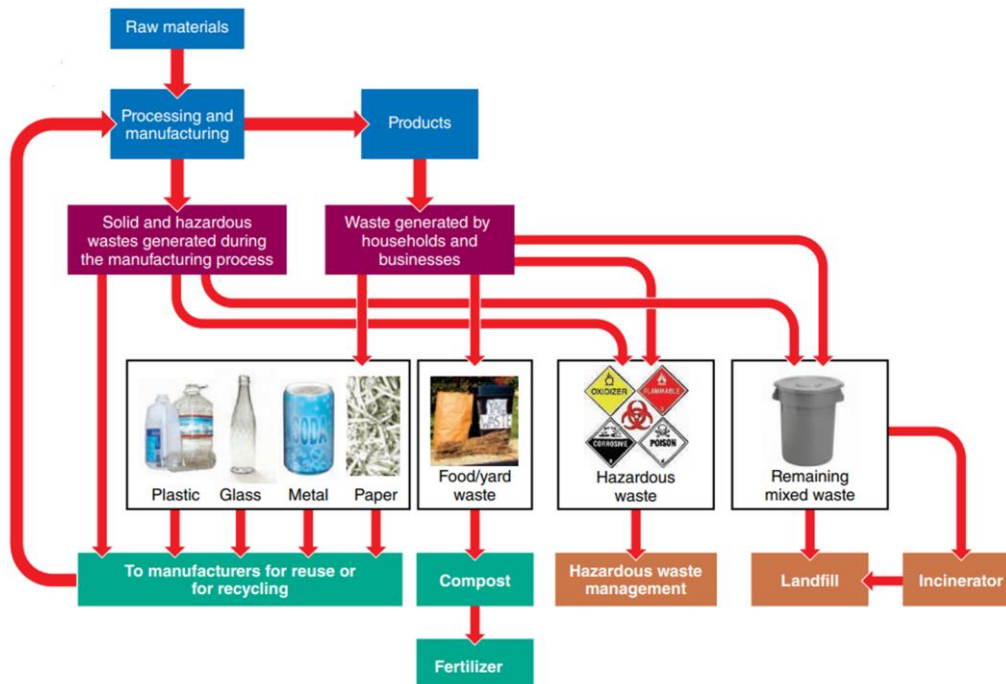
Integrated waste management has three main priorities for waste reduction and elimination.



Priorities to deal with waste according to Integrated Solid Waste Management. Reprinted from Fig 21-7, Living in the Environment, Tyler and Spoolman

It has three main priorities, which are summarized as follows:

1. The first priority is *control of pollution and waste prevention*. This involves changing industrial processes to eliminate the use of harmful chemicals. Swapping a more harmful product with a safer alternative; or, making products that last longer, and can be recycled, reused or are easy to repair.
2. The second priority is *control of secondary pollution and waste prevention*. This focuses mainly on the three Rs, i.e., reduce, reuse and recycle.
3. The third priority is *waste management*. This is adopted in cases where waste cannot be avoided, charting out safe methods to eliminate and dispose of hazardous waste, so that it does not cause further environmental pollution. This flow chart helps you understand the entire process of IWM. It lists out ways to deal with the different kinds of waste. E.g., plastic, glass, metal and paper can be reused or recycled, whereas, food waste or garden waste can be converted into compost. On the other hand, hazardous waste such as harmful chemicals and metals are dealt with in a different manner. They are either disposed off into sanitary landfills, so that it is burnt up in an incinerator.



Flowchart depicting the various approaches to dealing with solid waste. Reprinted from Fig 21-6, Living in the Environment, Tyler and Spoolman

Waste Disposal

Waste disposal refers to methods adopted in order to safely dispose off with waste in order to avoid further pollution and damage to the environment. Waste disposal can be done by either burning or burying solid waste. They can be classified into the following methods. Landfills: either open dumps or sanitary landfills, biocomposting, incineration

Open dumps

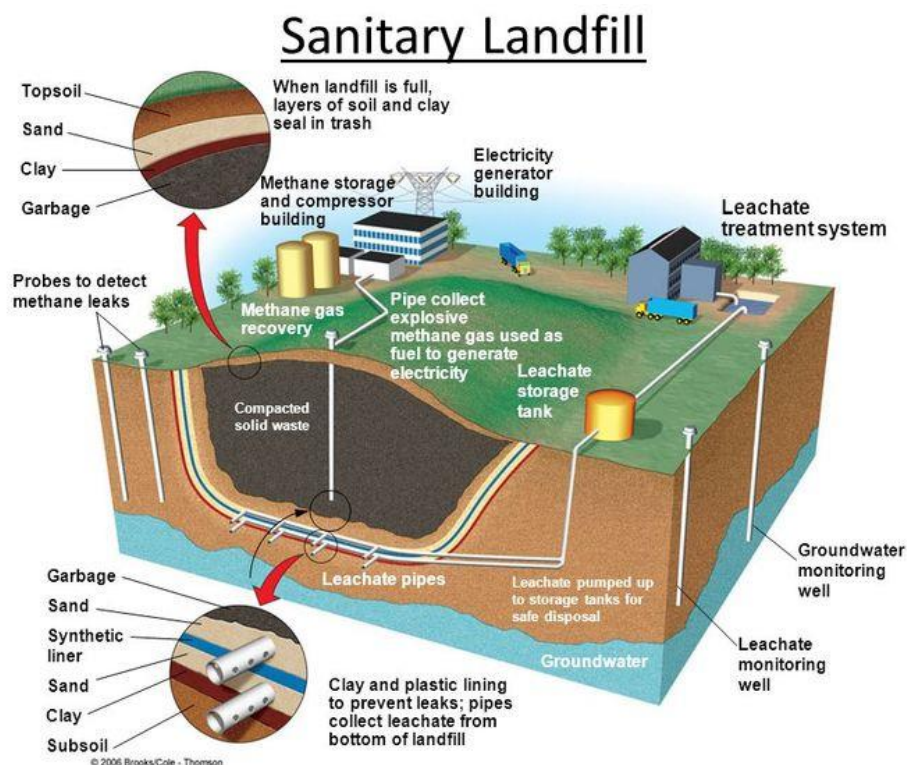
Open dumps are essentially fields or holes in the ground where garbage is deposited and sometimes burned. They are rare in more-developed countries, but are widely used near major cities in many less-developed countries, lacking a proper waste disposal system.



Sanitary landfills

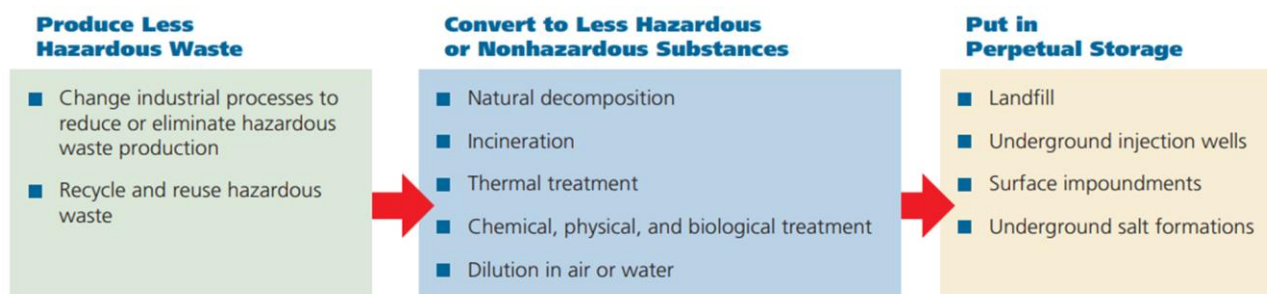
Sanitary landfills are a method of waste disposal in which solid wastes are spread out in thin layers, compacted and covered daily with a fresh layer of clay or plastic foam, which keeps the material dry and reduces leakage of contaminated water from the landfill. This covering also reduces the risk of fire, decreases odour and reduces accessibility to vermin. The bottom and sides are also lined with strong double liners and containment systems that collect liquid leaching (leachate) from them. Some are equipped with systems for collecting and burning methane, which is produced when wastes decompose in the absence of oxygen.

Sanitary landfills have their set of advantages and disadvantages. They are commonly used because they have low operating costs and can handle large amounts of waste. The filled land can also be used for other purposes. Certain disadvantages associated with sanitary landfills are: they can release greenhouse gases such as methane and carbon dioxide. They can contaminate the groundwater.



Disposal of hazardous waste

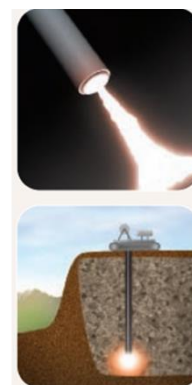
IWM suggests three priority levels in dealing with hazardous waste, namely, produce less; convert as much of it as possible to less hazardous substances; and put the rest in long-term, safe storage. The top priority should be pollution prevention and waste reduction. With this approach, industries try to find substitutes for toxic or hazardous materials, reuse or recycle the hazardous materials within industrial processes, or use them as raw materials for making other products.



Priorities to deal with hazardous waste according to Integrated Solid Waste Management. Reprinted from Fig 21-17, Living in the Environment, Tyler and Spoolman

Let us look at some of the methods to detoxify or dispose hazardous waste briefly. Some of the detoxification methods include: Physical methods, Chemical methods, Nanomagnets, Biological methods, Incineration and the Plasma arc torch method.

1. **Physical methods** for detoxifying hazardous wastes include using charcoal or resins to filter out harmful solids, distilling liquid wastes to separate out harmful chemicals, and precipitating, or allowing natural processes to separate, such chemicals from solution.
2. **Chemical methods** are used to convert hazardous chemicals to harmless or less harmful chemicals through chemical reactions. For example, cyclodextrin (a type of sugar made from cornstarch) is used to remove toxic materials such as solvents and pesticides from contaminated soil and groundwater. Cyclodextrin acts like a sponge picking up chemicals from the soil.
3. **Nanomagnets** are magnetic nanoparticles coated with certain compounds that can remove various pollutants from water. E.g., magnetic nanoparticles coated with chitosan, derived from the exoskeletons of shrimps and crabs, are used to remove oil and other organic pollutants from contaminated water. Magnetic fields are used to remove the pollutant-coated nanomagnets. The pollutants can then be separated out and disposed of or recycled, and the magnetic nanoparticles can be reused.
4. **Biological methods** are used for treatment of hazardous waste. Some of the methods used are bioremediation and phytoremediation, bioremediation utilizes Bacteria and enzymes help destroy toxic/hazardous substances, or convert them to harmless compounds. Phytoremediation involves using natural or genetically engineered plants to absorb, filter, and remove contaminants from polluted soil and water.



Plasma-Arc Torch Method

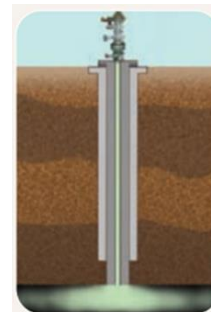
5. **Plasma arc torch** breaks them down at very high temperatures. Plasma

can decompose liquid or solid hazardous waste to gas consisting mostly of carbon monoxide (CO) and hydrogen (H₂) and a molten, glassy, solid material can be used to encapsulate toxic metals and keep them from leaching into groundwater. This method is not widely used due to its high cost.

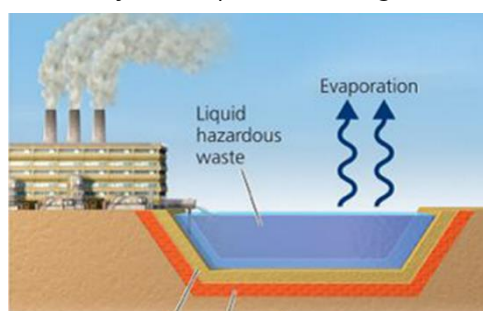
Storage of hazardous waste

Storage of hazardous waste such as burial or long-term storage should be considered as the third and final resort, after the first two priorities have been exhausted. The most commonly used disposal methods are deep-well disposal and surface impoundments.

1. In **deep-well disposal**, liquid hazardous wastes are pumped under pressure through a pipe into dry, porous rock formations far beneath aquifers that are tapped for drinking and irrigation water. However, this fairly cheap, out-of-sight and out-of-mind approach presents some



Deep-well disposal



Surface Impoundments

problems. There are a limited number of such sites and limited space within them. Sometimes the wastes can leak into groundwater from the well shaft or migrate into groundwater in unexpected ways.

2. **Surface impoundments** are lined ponds, pits, or lagoons in which liquid hazardous wastes are stored as the water evaporates, the waste settles and becomes more concentrated. But using no liner, using leaking single liners, and failing to use double liners can allow such wastes to percolate into the groundwater, and because these impoundments are not covered, volatile harmful chemicals can evaporate into the air.
3. **Secure hazardous waste landfills:** are used for waste that cannot be destroyed, detoxified or safely buried. In such cases, they are put into metal drums or containers that are buried in salt mines or bedrock caverns, where they can be inspected on a regular basis, and retrieved, if necessary.

Steps to reduce waste as an individual

- Follow the three Rs, i.e., reduce, reuse and recycle.
- Try to avoid packaging material whenever possible when you buy a particular item.
- Rent, borrow or barter goods and services when you can buy second-hand, and donate or sell unused items.
- Buy things that are reusable, recyclable or compostable, and be sure to reuse, recycle and compost them.
- Avoid disposables and throwaway paper and plastic cups, eating utensils and other disposable items when reusable versions are available.
- Use email or text-messaging in place of conventional paper email. Read newspapers and magazines online and read e-books.
- Buy products in bulk or concentrated form whenever possible.

Steps from the government to improve recycling and waste reduction

- Increase in subsidies and tax breaks for reuse and recycling materials (positive incentive).

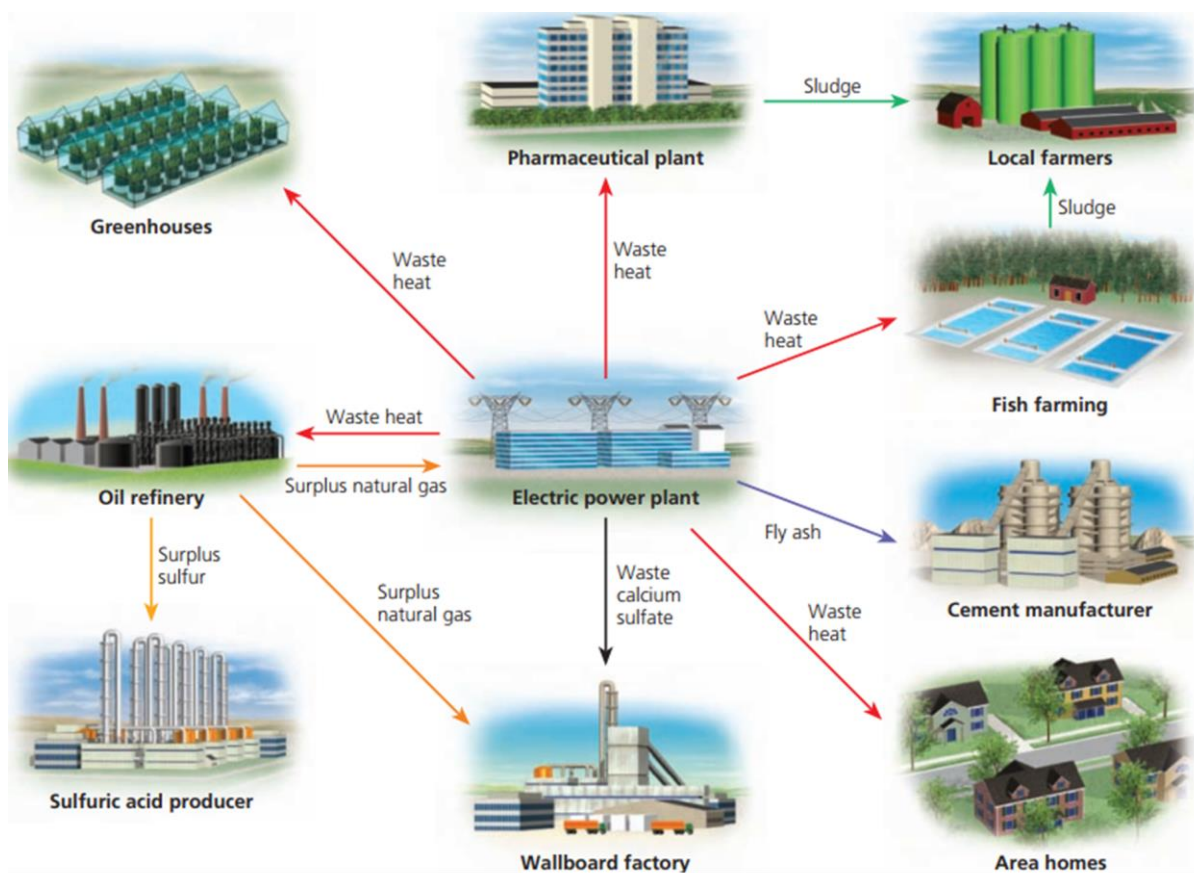
- Decrease subsidies and tax breaks for producing items from virgin resources (negative incentive).
- Fee-per-bag waste collection systems.
- Encourage purchases of recycled products.
- Pass laws requiring companies to take back and recycle/reuse packaging and electronic waste discarded by consumers.

Case study: Industrial Ecosystems: Biomimicry

Biomimicry is the science and art of discovering and using natural principles to help solve human problems. For example, scientists have studied termite mounds to learn how to cool buildings naturally. One way for industries to mimic nature is to reuse or recycle most of the minerals and chemicals they use, instead of burying or burning them or shipping them somewhere. Another way for industries to mimic nature would be to interact through resource exchange webs in which the wastes of one manufacturer become the raw materials for another—similar to food webs in natural ecosystems.

Industrial ecosystem: Kalundborg, Denmark

In Kalundborg, an electric power plant and nearby industries, farms, and homes are collaborating to save money and to reduce their outputs of waste and pollution, within what is called an eco-industrial park, or industrial ecosystem. They exchange waste outputs and convert them into resources. This cuts pollution and waste and reduces the flow of non-renewable mineral and energy resources through the local economy.



This industrial ecosystem in Kalundborg, Denmark, reduces waste production by mimicking a natural ecosystem's food web. Reprinted from Fig 21-26, *Living in the Environment*, Tyler and Spoolman

Biomimicry also encourages companies to come up with new, environmentally beneficial, and less resource-intensive chemicals, processes, and products that they can sell worldwide. In addition, these companies convey a better image to consumers based on actual results rather than public relations campaigns. Biomimicry involves two major steps.

- The first is to observe certain changes in nature and to study how natural systems have responded to such changing conditions over many millions of years.
- The second step is to try to copy or adapt these responses within human systems in order to help us deal with various environmental challenges. In the case of solid and hazardous wastes, the food web serves as a natural model for responding to the growing problem of these wastes.